Notice that the definition of $\widetilde{X}_\rho$ makes sense whenever we are given an action ρ of $\pi_1(X,x_0)$ on a set F. There is a natural projection $\widetilde{X}_\rho \to X$ sending $([\gamma],\widetilde{x}_0)$ to $\gamma(1)$, and this is a covering space since if $U \subset X$ is an open set over which the universal cover $\widetilde{X}_0$ is a product $U\times\pi_1(X,x_0)$, then the identifications defining $\widetilde{X}_\rho$ simply collapse $U\times\pi_1(X,x_0)\times F$ to $U\times F$.

Returning to our given covering space $\widetilde{X}\to X$ with associated action ρ, the map $\widetilde{X}_\rho \to \widetilde{X}$ induced by h is a bijection and therefore a homeomorphism since h was a local homeomorphism. Since this homeomorphism $\widetilde{X}_\rho \to \widetilde{X}$ takes each fiber of $\widetilde{X}_\rho$ to the corresponding fiber of $\widetilde{X}$, it is an isomorphism of covering spaces.

If two covering spaces $p_1:\widetilde{X}_1\to X$ and $p_2:\widetilde{X}_2\to X$ are isomorphic, one may ask how the corresponding actions of $\pi_1(X,x_0)$ on the fibers F_1 and F_2 over x_0 are related. An isomorphism $h:\widetilde{X}_1\to\widetilde{X}_2$ restricts to a bijection $F_1\to F_2$, and evidently $L_\gamma(h(\widetilde{x}_0)) = h(L_\gamma(\widetilde{x}_0))$. Using the less cumbersome notation $\gamma\widetilde{x}_0$ for $L_\gamma(\widetilde{x}_0)$, this relation can be written more concisely as $\gamma h(\widetilde{x}_0) = h(\gamma\widetilde{x}_0)$. A bijection $F_1\to F_2$ with this property is what one would naturally call an *isomorphism of sets with $\pi_1(X,x_0)$ action*. Thus isomorphic covering spaces have isomorphic actions on fibers. The converse is also true, and easy to prove. One just observes that for isomorphic actions ρ_1 and ρ_2, an isomorphism $h:F_1\to F_2$ induces a map $\widetilde{X}_{\rho_1}\to\widetilde{X}_{\rho_2}$ and h^{-1} induces a similar map in the opposite direction, such that the compositions of these two maps, in either order, are the identity.

This shows that n-sheeted covering spaces of X are classified by equivalence classes of homomorphisms $\pi_1(X,x_0)\to\Sigma_n$, where Σ_n is the symmetric group on n symbols and the equivalence relation identifies a homomorphism ρ with each of its conjugates $h^{-1}\rho h$ by elements $h\in\Sigma_n$. The study of the various homomorphisms from a given group to Σ_n is a very classical topic in group theory, so we see that this algebraic question has a nice geometric interpretation.

Deck Transformations and Group Actions

For a covering space $p:\widetilde{X}\to X$ the isomorphisms $\widetilde{X}\to\widetilde{X}$ are called **deck transformations** or **covering transformations**. These form a group $G(\widetilde{X})$ under composition. For example, for the covering space $p:\mathbb{R}\to S^1$ projecting a vertical helix onto a circle, the deck transformations are the vertical translations taking the helix onto itself, so $G(\widetilde{X})\approx\mathbb{Z}$ in this case. For the n-sheeted covering space $S^1\to S^1$, $z\mapsto z^n$, the deck transformations are the rotations of S^1 through angles that are multiples of $2\pi/n$, so $G(\widetilde{X})=\mathbb{Z}_n$.

By the unique lifting property, a deck transformation is completely determined by where it sends a single point, assuming $\widetilde{X}$ is path-connected. In particular, only the identity deck transformation can fix a point of $\widetilde{X}$.

A covering space $p:\widetilde{X}\to X$ is called **normal** if for each $x\in X$ and each pair of lifts $\widetilde{x},\widetilde{x}'$ of x there is a deck transformation taking $\widetilde{x}$ to $\widetilde{x}'$. For example, the covering